

LAB REPORT – PART B: HOMEWORK EXERCISES

STUDENT INFORMATION:

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Class: Web Application Development_S1_2025-26_G02_lab01

COMPLETED EXERCISES:

☒ Exercise 5: Authentication Filter

☒ Exercise 6: Admin Authorization Filter

☒ Exercise 7: Role-Based UI

☐ Exercise 8: Pagination

☐ Bonus 1: Export Excel

OVERVIEW:

This lab focused on implementing authentication, role-based authorization, and dynamic UI rendering in a Java web application using Jakarta Servlets. Filters were used to secure protected pages and enforce admin-only operations, while JSP was updated to display user-specific controls and messages.

EXERCISE 5 – AUTHENTICATION FILTER (AuthFilter.java):

- Intercepts all HTTP requests using the `@WebFilter` annotation.
- Maintains a list of public URLs and static resources (login, logout, CSS, JS, images) that bypass authentication.
- Checks session for logged-in user; redirects to login page if unauthenticated.
- Logs initialization and destruction events for traceability.
- Verified behavior by accessing protected pages and confirming redirect to login when not authenticated.

EXERCISE 6 – ADMIN AUTHORIZATION FILTER (AdminFilter.java):

- Intercepts requests to `/student` URL to check for admin-only actions.
- Defined `ADMIN_ACTIONS` array for operations restricted to admins: new, insert, edit, update, delete.
- Retrieves user from session and verifies admin role; redirects with error message if access denied.
- Non-admin users can view list but cannot perform restricted actions.

- Logs initialization and destruction events for maintainability.
- Verified by testing with both admin and regular users; direct URL access to restricted actions properly blocked.

EXERCISE 7 – ROLE-BASED UI (student-list.jsp):

- Added a navigation bar displaying logged-in user name and role badge.
- Display of "Add New Student" and Edit/Delete buttons conditional on user role (admin only).
- Error messages from AdminFilter displayed in the interface when access is denied.
- Table layout dynamically adjusts based on admin privileges; ensures non-admin users cannot interact with restricted actions.
- Enhanced styling for clarity and user experience, including responsive design, hover effects, and distinct success/error message formatting.

MVC COMPONENTS:

- Filters: AuthFilter.java, AdminFilter.java
- Controller: StudentController.java
- Views: student-list.jsp, student-form.jsp
- Model: User.java

FEATURES IMPLEMENTED:

- Request interception and session-based authentication
- Role-based access control for admin operations
- Dynamic UI rendering based on user role
- Display of error messages for unauthorized access attempts
- Logging of filter lifecycle events for debugging

KNOWN ISSUES:

- No functional issues detected
- Possible improvement: externalize public URL patterns and admin actions for easier maintenance

EXTRA FEATURES:

- UI styling improvements for better readability and user experience

TIME SPENT: 3 hours

*Screenshots

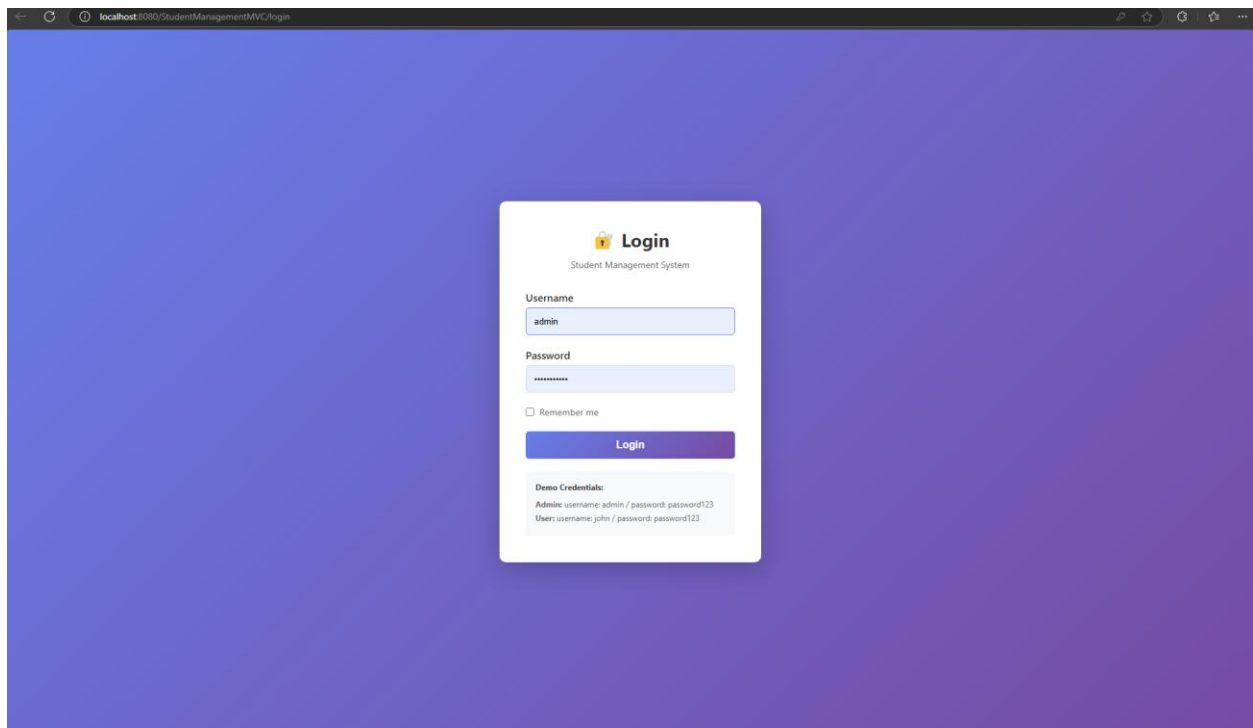


Figure 1: Log in view

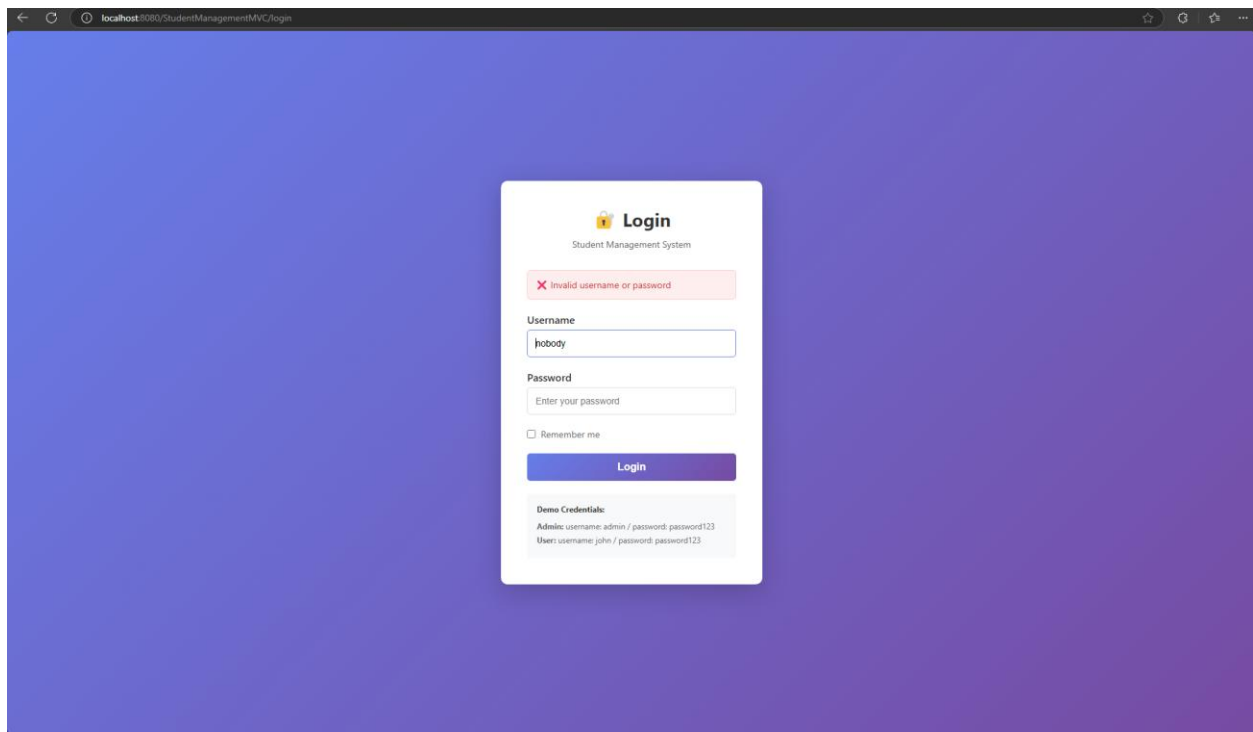


Figure 2: Log in with wrong credentials

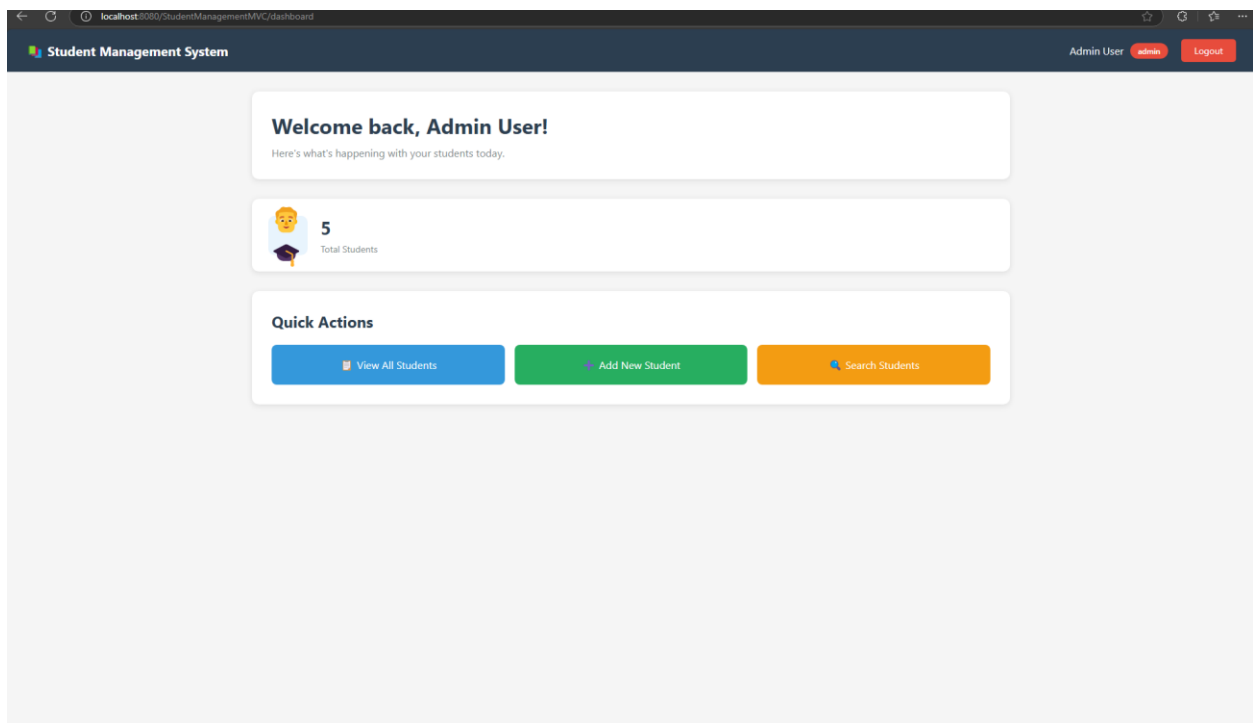


Figure 3: Dashboard view

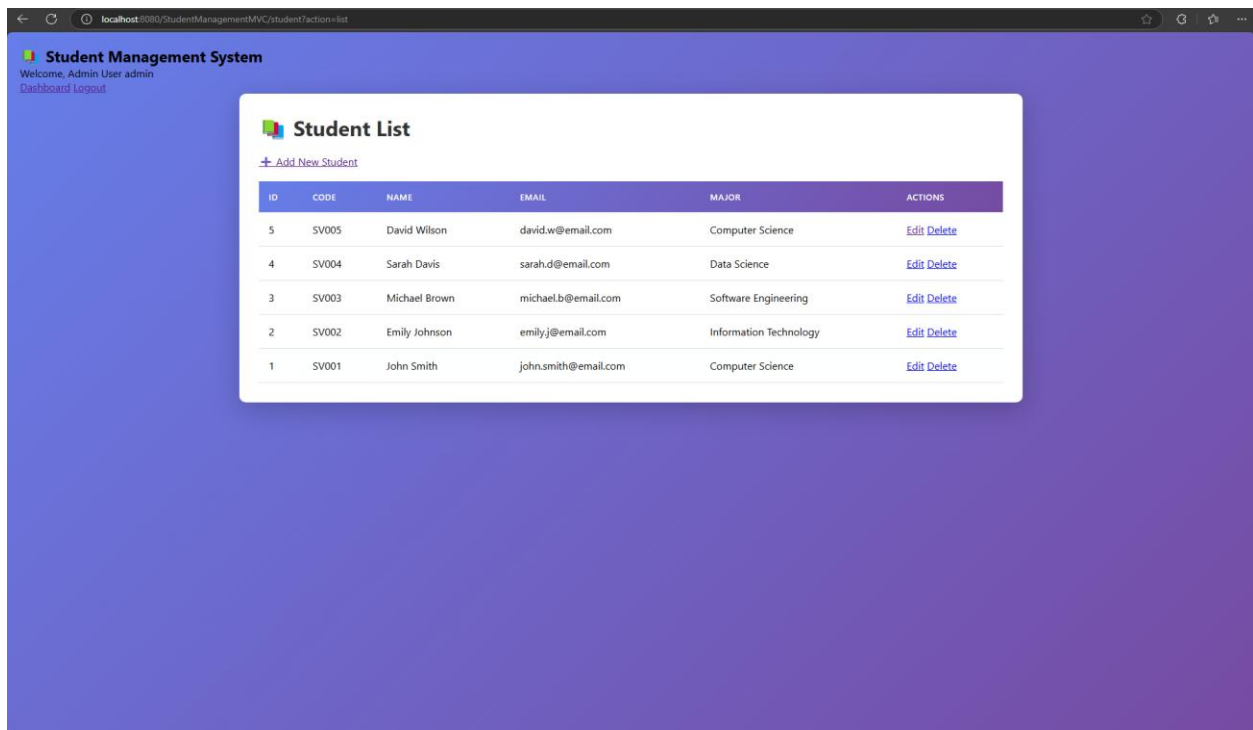


Figure 4: Admin student list

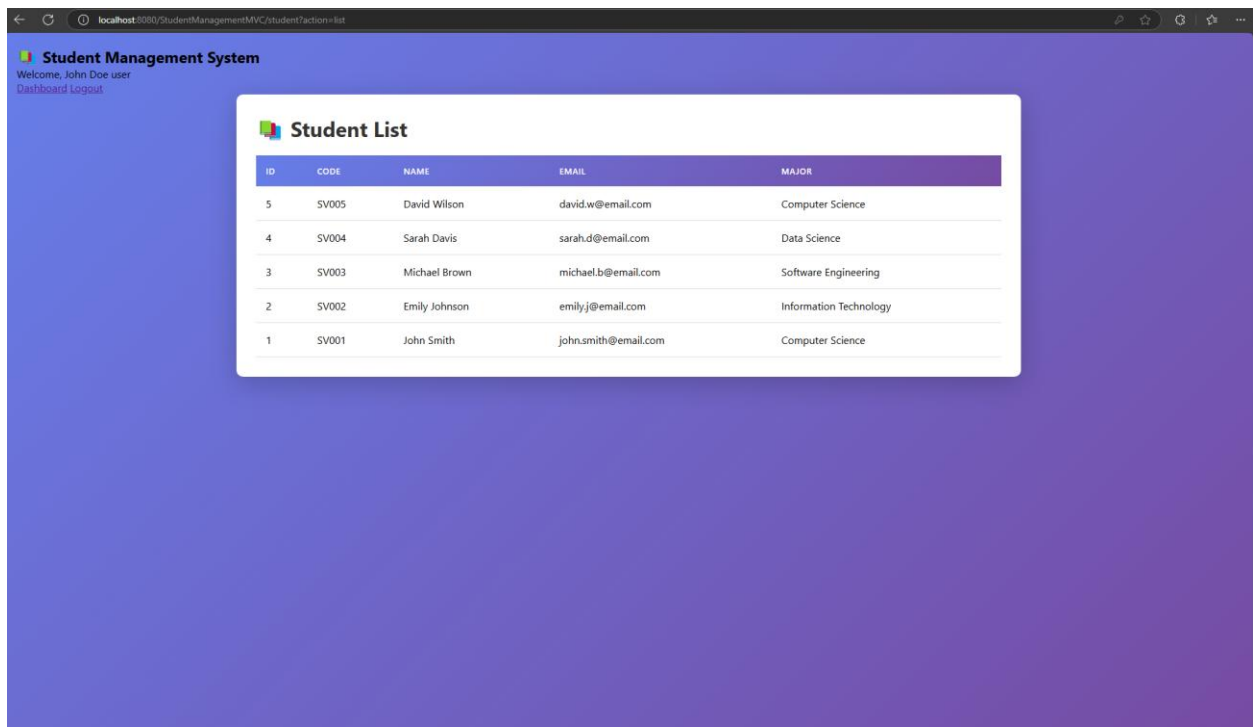


Figure 5: Normal user student list

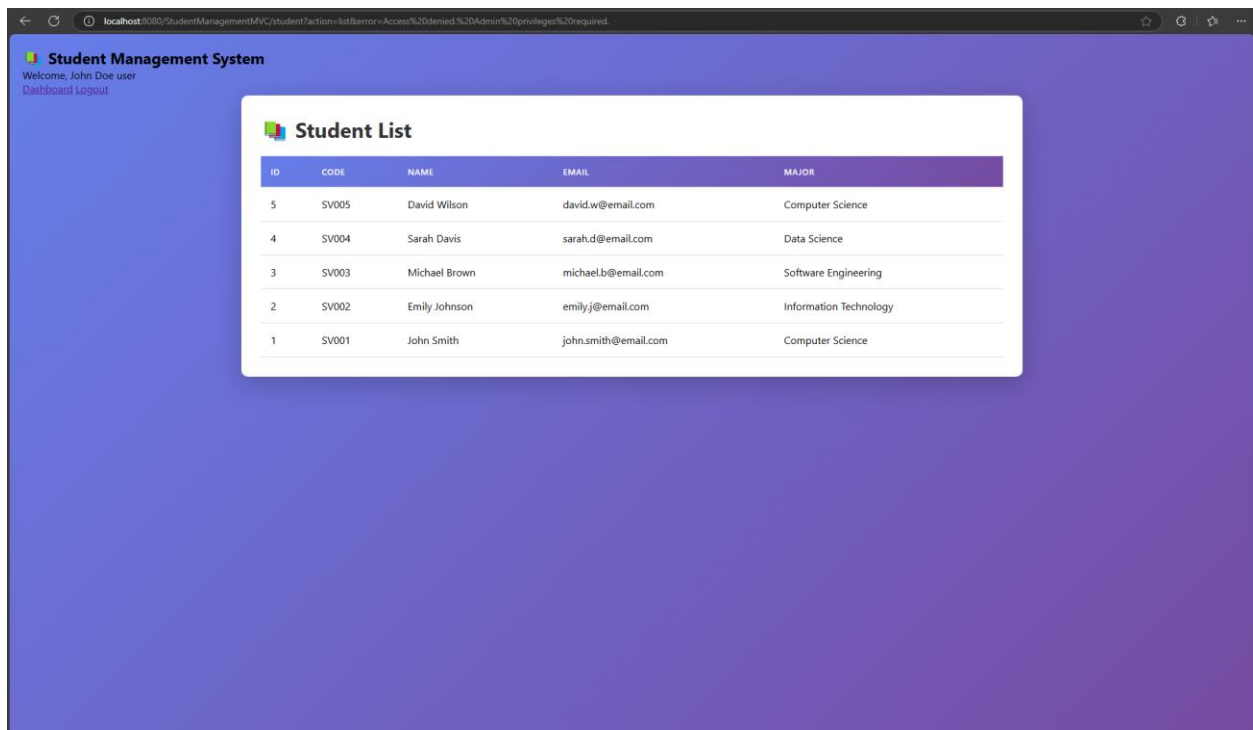


Figure 6: Error message in url warning access denied

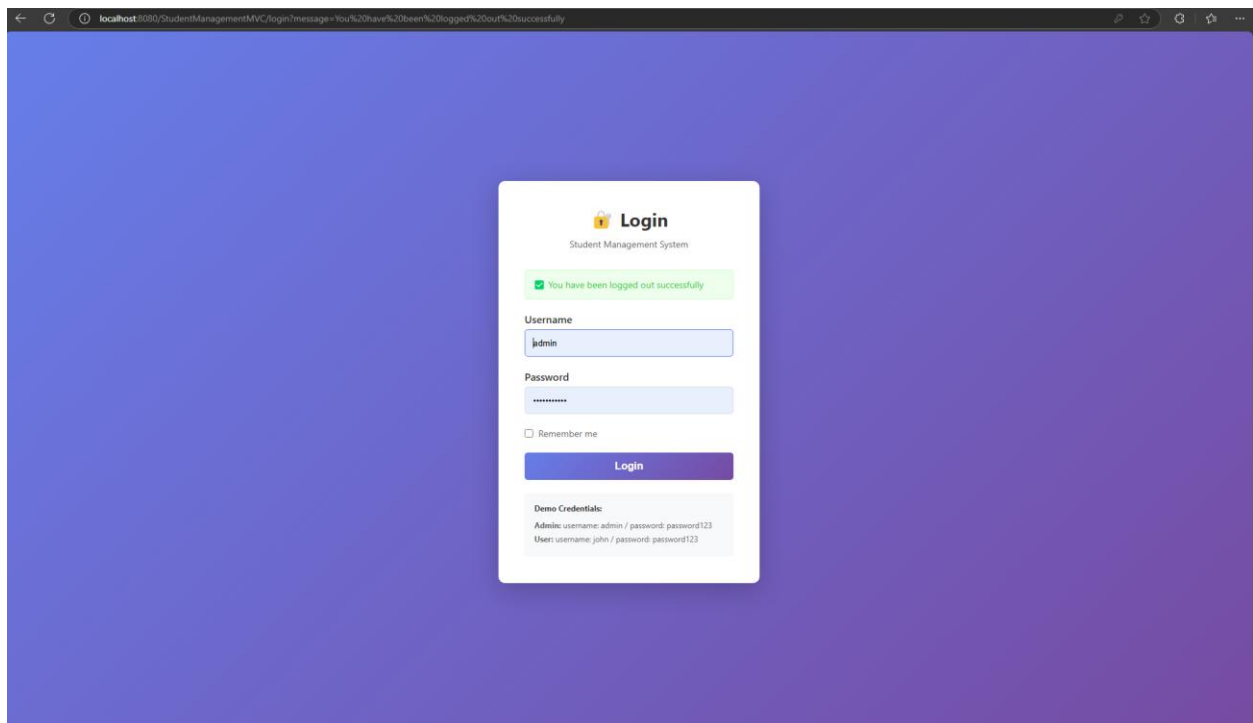


Figure 7: Log out successfully